5.3: The Other Trigonometric Functions

Definition.

$$\sec(\theta) = \frac{1}{\cos(\theta)} \qquad \tan(\theta) = \frac{\sin(\theta)}{\cos(\theta)}$$

$$\csc(\theta) = \frac{1}{\sin(\theta)} \qquad \cot(\theta) = \frac{\cos(\theta)}{\sin(\theta)}$$

Example. Simplify the following:

$$\frac{\sec(\theta)}{\tan(\theta)} = \frac{1/\cos\theta}{\sin\theta/\cos\theta}$$

$$= \frac{1}{\cos\theta} \cdot \frac{\cos\theta}{\sin\theta}$$

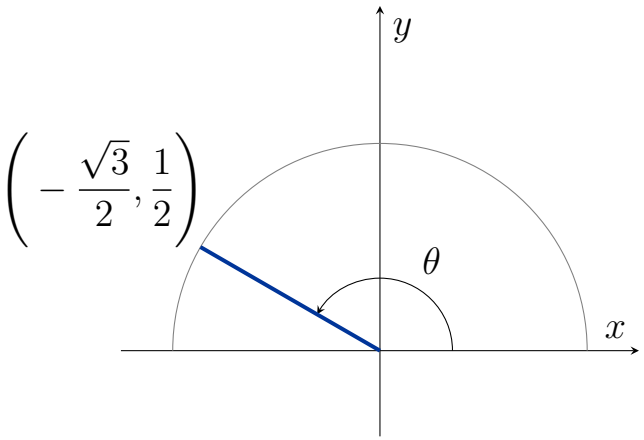
$$= \frac{1}{\sin\theta}$$

$$= \boxed{\csc\theta}$$

$$\tan(\theta)\cot(\theta) = \frac{\sin\theta}{\cos\theta} \cdot \frac{\cos\theta}{\sin\theta}$$

$$= 1 \quad \left(\theta \neq \frac{n\pi}{2}\right)$$

Example. Consider the point $\left(-\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$ on the unit circle. Find the value of the six trigonometric functions:



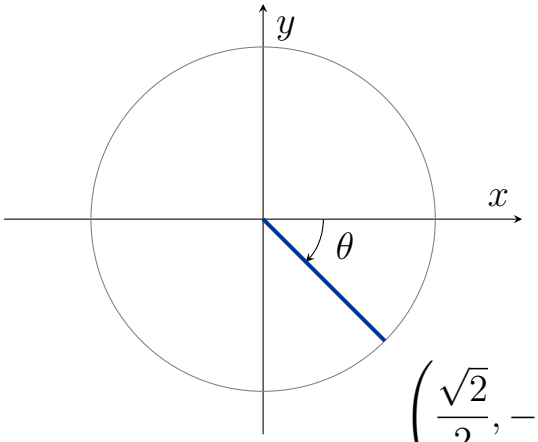
$\cos(\theta) = \boxed{-\frac{\sqrt{3}}{2}}$

$\sin(\theta) = \boxed{\frac{1}{2}}$

$\sec(\theta) = \frac{1}{\cos\theta} = \frac{1}{-\sqrt{3}/2} = -\frac{2}{\sqrt{3}} = \boxed{-\frac{2\sqrt{3}}{3}}$ $\csc(\theta) = \frac{1}{\sin\theta} = \frac{1}{1/2} = \boxed{2}$

$\tan(\theta) = \frac{\sin\theta}{\cos\theta} = \frac{1/2}{-\sqrt{3}/2} = -\frac{1}{\sqrt{3}} = \boxed{-\frac{\sqrt{3}}{3}}$ $\cot(\theta) = \frac{\cos\theta}{\sin\theta} = \frac{-\sqrt{3}/2}{1/2} = \boxed{-\sqrt{3}}$

Example. Consider the point $\left(\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right)$ on the unit circle. Find the value of the six trigonometric functions:



$\left(\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right)$

$\cos(\theta) = \boxed{\frac{\sqrt{2}}{2}}$

$\sin(\theta) = \boxed{-\frac{\sqrt{2}}{2}}$

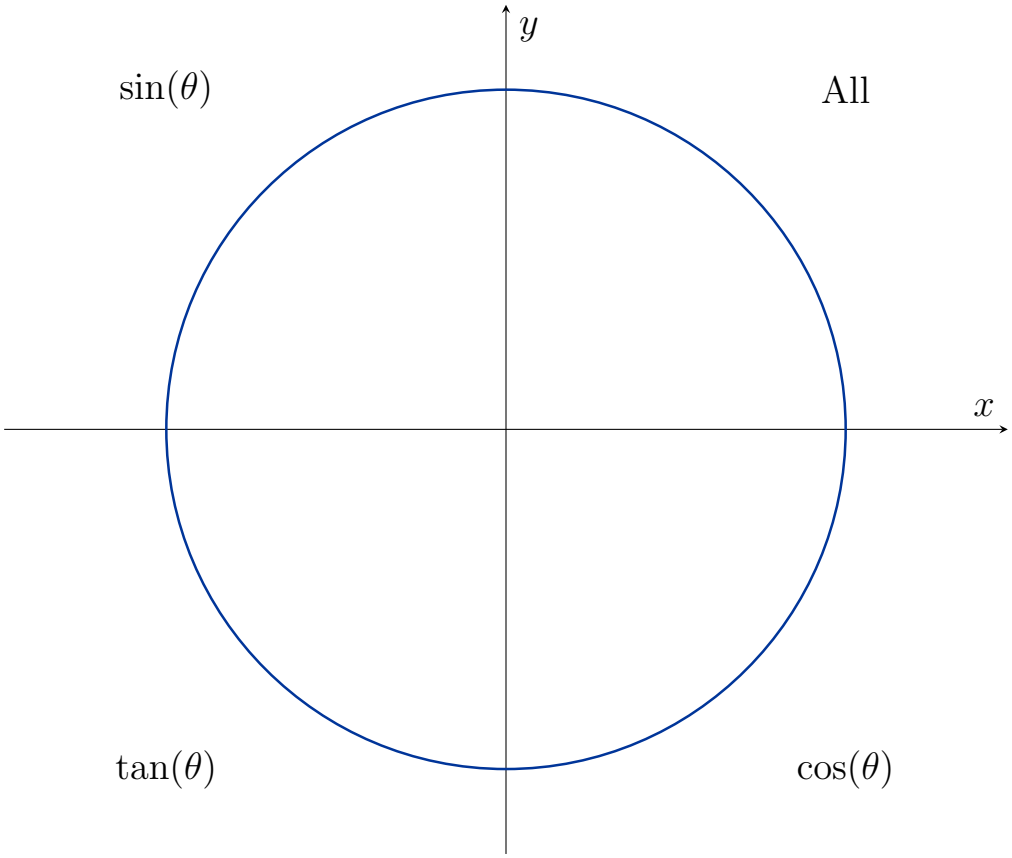
$\sec(\theta) = \frac{1}{\cos\theta} = \frac{1}{\sqrt{2}/2} = \frac{2}{\sqrt{2}} = \boxed{\sqrt{2}}$

$\csc(\theta) = \frac{1}{\sin\theta} = \frac{1}{-\sqrt{2}/2} = -\frac{2}{\sqrt{2}} = \boxed{-\sqrt{2}}$

$\tan(\theta) = \frac{\sin\theta}{\cos\theta} = \frac{-\sqrt{2}/2}{\sqrt{2}/2} = \boxed{-1}$

$\cot(\theta) = \frac{\cos\theta}{\sin\theta} = \frac{\sqrt{2}/2}{-\sqrt{2}/2} = \boxed{-1}$

Note: The sign of each trig. function is determined by the quadrant



Example. Find the value of the six trigonometric functions when

$\theta = \frac{\pi}{6}$

$\cos\theta = \frac{\sqrt{3}}{2}$ $\quad \sin\theta = \frac{1}{2}$

$\sec\theta = \frac{2\sqrt{3}}{3}$ $\quad \csc\theta = 2$

$\cot\theta = \sqrt{3}$ $\quad \tan\theta = \frac{\sqrt{3}}{3}$

$\theta = \frac{\pi}{3}$

$\cos\theta = \frac{1}{2}$ $\quad \sin\theta = \frac{\sqrt{3}}{2}$

$\sec\theta = 2$ $\quad \csc\theta = \frac{2\sqrt{3}}{3}$

$\cot\theta = \frac{\sqrt{3}}{3}$ $\quad \tan\theta = \sqrt{3}$

$\theta = -\frac{5\pi}{6}$

$\cos\left(-\frac{5\pi}{6}\right) = \cos\left(\frac{5\pi}{6}\right) = -\frac{\sqrt{3}}{2}$ $\quad \sin\left(-\frac{5\pi}{6}\right) = -\sin\left(\frac{5\pi}{6}\right) = -\frac{1}{2}$

$\sec\theta = -\frac{2\sqrt{3}}{3}$ $\quad \csc\theta = -2$

$\cot\theta = \sqrt{3}$ $\quad \tan\theta = \frac{\sqrt{3}}{3}$

$\theta = -\frac{7\pi}{4}$

$\cos\left(-\frac{7\pi}{4}\right) = \cos\left(\frac{7\pi}{4}\right) = -\frac{\sqrt{2}}{2}$ $\quad \sin\left(-\frac{7\pi}{4}\right) = -\sin\left(\frac{7\pi}{4}\right) = -\frac{\sqrt{2}}{2}$

$\sec\theta = -\sqrt{2}$ $\quad \csc\theta = -\sqrt{2}$

$\cot\theta = 1$ $\quad \tan\theta = 1$

Recall that $\cos(\theta)$ is even, and $\sin(\theta)$ is odd:

$$\cos(\theta) = \cos(-\theta) \qquad -\sin(\theta) = \sin(-\theta)$$

Thus,

$$\sec(\theta) = \sec(-\theta) \qquad -\csc(\theta) = \csc(-\theta)$$

$$-\tan(\theta) = \tan(-\theta) \qquad -\cot(\theta) = \cot(-\theta)$$

Example. If $\sec(\theta) = 2$, what is $\sec(-\theta)$?

$\sec\theta = 2 \implies \sec(-\theta) = \boxed{2}$

Example. If $\cot(\theta) = \sqrt{3}$, what is $\cot(-\theta)$?

$\cot(\theta) = \sqrt{3} \implies \cot(-\theta) = \boxed{-\sqrt{3}}$

Alternate forms of the Pythagorean Theorem

Recall the Pythagorean Identity:

$$\cos^2(\theta) + \sin^2(\theta) = 1$$

We now define alternate forms:

$$1 + \tan^2(\theta) = \sec^2(\theta)$$

$$\cot^2(\theta) + 1 = \csc^2(\theta)$$

Example. If $\tan(\theta) = -\sqrt{3}$, and $\pi < \theta < 2\pi$, find $\sec(\theta)$.

$1 + \tan^2\theta = \sec^2\theta$

$1 + (-\sqrt{3})^2 = \sec^2\theta$

$4 = \sec^2\theta$

$\pm 2 = \sec\theta$

4th quad

$\Rightarrow \boxed{\sec\theta = -2}$

Alt.

$\tan\theta = -\sqrt{3} = \frac{\sqrt{3}/2}{-1/2}$

$\Rightarrow \sin\theta = \sqrt{3}/2$

$\cos\theta = -1/2$

$\Rightarrow \sec\theta = -2$

Example. If $\csc(\theta) = -\frac{2\sqrt{3}}{3}$, and $\pi < \theta < \frac{3\pi}{2}$, find $\cot(\theta)$.

$\cot^2\theta + 1 = \csc^2\theta$

$\cot^2\theta + 1 = \left(-\frac{2\sqrt{3}}{3}\right)^2$

$\cot^2\theta = \frac{4}{3} - 1$

$\cot\theta = \pm\frac{\sqrt{3}}{3}$

3rd quad

$\Rightarrow \boxed{\cot\theta = \frac{\sqrt{3}}{3}}$

Alt

$\csc\theta = -\frac{2\sqrt{3}}{3} = -\frac{2}{\sqrt{3}}$

$\Rightarrow \sin\theta = -\frac{\sqrt{3}}{2}$

$\cos\theta = -1/2$

$\Rightarrow \cot\theta = \frac{-1/2}{-\sqrt{3}/2} = \frac{1}{\sqrt{3}} = \frac{\sqrt{3}}{3}$

Definition. (Period of a Function)
The **period** P of a repeating function f is the number representing the interval such that $f(x+P) = f(x)$ for any value of x.

The period for cosine, sine, secant, and cosecant is 2π.

The period for tangent and cotangent is π.

Example. If $\sin(\theta) = \frac{1}{5}$, find $\sin(\theta + 2\pi)$ and $\sin(\theta - 2\pi)$.

$$\sin(\theta + 2\pi) = \frac{1}{5}$$

$$\sin(\theta - 2\pi) = \frac{1}{5}$$

Example. If $\tan(\theta) = \frac{\sqrt{3}}{3}$, find $\tan(\theta + \pi)$ and $\cot(\theta + \pi)$.

$$\tan(\theta + \pi) = \frac{\sqrt{3}}{3}$$

$$\cot(\theta - \pi) = \cot(\theta) = \sqrt{3}$$

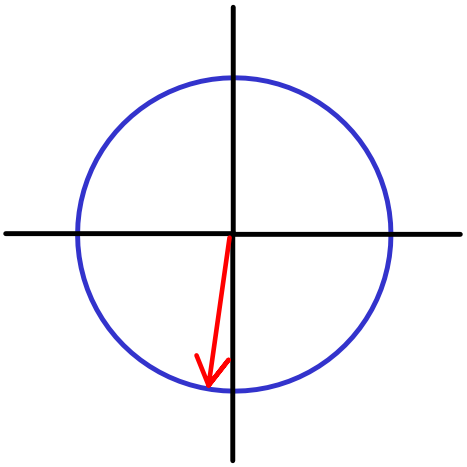
Example. Find the value of all six trigonometric functions and identify which quadrant θ is in:

$$\cos(\theta) = \frac{5}{13} \text{ and } \sin(\theta) = -\frac{12}{13}$$

3rd Quad

$\sec\theta = \frac{13}{5}$ $\quad$ $\csc\theta = -\frac{13}{12}$

$\cot\theta = -\frac{5}{12}$ $\quad$ $\tan\theta = -\frac{12}{5}$



$$\csc(\theta) = -\sqrt{2} \text{ and } \tan(\theta) = -1$$

$\sin\theta = -\frac{\sqrt{2}}{2}$

$\sin\theta = -\cos\theta$

$\Rightarrow \cos\theta = \frac{\sqrt{2}}{2}$

$\Rightarrow \sec\theta = \sqrt{2}$

$\cot\theta = -1$

4th Quad

